AI EFFECT

AI EFFECT Solution Studio Production Operations Guide

Infrastructure ownership security cost management and operational procedures

This guide provides the IT team with the information required to operate, secure, monitor, back up, recover, and budget for the production AI EFFECT Solution Studio and its four country node use cases. It records the verified production state as of 4 October 2026 and identifies the controls that still require implementation.

Document field	Value
Audience	AI EFFECT IT operations, infrastructure and security teams
Environment	Production
Primary service	https://platform.ai-effect.eu/
Secondary hostname	https://portal.renewenergy.io/
Production host	evpa-uat at 212.47.78.250
Hosting provider	Contabo GmbH
Document version	1.0
Status date	4 October 2026
Git branch	feature/production-ui-polish-v1
Recorded commit	ddcfc3b

Classification Internal operational documentation. Do not attach credentials, passwords, private keys, tokens, or environment-file contents.

Contents

1. Executive summary
2. Production architecture
3. Infrastructure inventory
4. Country node services and data
5. Capacity and storage
6. Expenses and budget planning
7. Security controls
8. Backup and recovery
9. Monitoring and alerting
10. Routine operations and change management
11. Incident response and rollback
12. Risks and prioritized action plan
13. Ownership and access management
14. Appendix A Operational commands
15. Appendix B Reference sources

Document use

Sections 2 through 6 explain what is running and what it costs. Sections 7 through 11 define the required operating controls. Section 12 should be used as the initial remediation backlog. Appendix A contains read-only commands for routine diagnosis.

1 Executive summary

The AI EFFECT Solution Studio and all four country node use cases currently run as Docker containers on one shared Contabo VPS. The portal does not depend on a developer laptop. Browser traffic reaches Traefik over HTTPS, the Solution Studio communicates with a private API and WP3 orchestrator, and the orchestrator invokes country services over private Docker networks.

The server has sufficient capacity for the initial controlled release. It provides 8 virtual CPUs, 24 GB RAM and a 200 GB NVMe disk. At the time of measurement, 89 GB was used, 97 GB was available, approximately 4.2 GB of RAM was used across the whole server, and approximately 18 GB of RAM remained available.

The four use cases passed end-to-end production acceptance. Twelve country-node workflow tasks completed successfully and returned output references. The portal catalogue contains sixteen services, including the twelve accepted German, Portuguese, Danish and Dutch services.

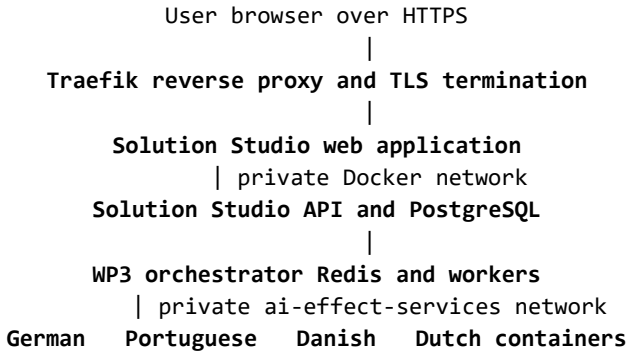
Immediate IT priorities

1. Create and test a nightly backup for the new Solution Studio PostgreSQL database and copy it off the VPS.
2. Add automatic restart policies and Docker health checks to the German containers and Portuguese ClickHouse.
3. Configure bounded Docker log rotation so a noisy service cannot fill the system disk.
4. Repair or stop the unrelated evpa-nginx-1 container, which is continuously restarting.
5. Implement external availability, disk, backup and container-health alerts before broad public promotion.

Capacity decision No immediate VPS upgrade is required. A dedicated AI EFFECT VPS should be considered when the system begins handling sustained public traffic or authoritative user data.

2 Production architecture

The production design separates the public web entry point from private application and country-node services. Only the reverse proxy should accept public web traffic for AI EFFECT.



The public portal is available at platform.ai-effect.eu and portal.renewenergy.io. Both hostnames resolve to 212.47.78.250 and use the same production application. Traefik manages TLS certificates and routing.

Network trust boundaries

Boundary	Current design	IT requirement
Internet to Traefik	HTTPS on ports 80 and 443 with automatic certificate handling	Monitor certificate renewal and redirect HTTP to HTTPS
Traefik to web	Private Docker network; web also binds to 127.0.0.1:13000	Do not publish the application port externally
Web to API	Private Docker network	Keep API database credentials server-side
API to WP3	Orchestrator bound to 127.0.0.1:18000 and private networks	Do not expose orchestrator or Redis publicly
WP3 to country nodes	External private Docker network named ai-effect-services	Country services must not publish host ports
Administrator access	SSH key authentication to the renew account	Use named individual keys and remove departed users promptly

3 Infrastructure inventory

Server specification

Item	Observed value	Operational note
Provider	Contabo GmbH	Confirmed from the public IP autonomous system
Host name	evpa-uat	Shared with PowerPal and legacy EVPA workloads
Operating system	Ubuntu 22.04	Maintain security updates and supported packages
Compute	8 vCPUs on AMD EPYC	Shared VPS CPU; country workloads may contend during heavy runs
Memory	23.47 GiB usable	Approximately 18 GiB available at the status snapshot
Disk	196 GiB formatted ext4	89 GiB used and 97 GiB free at the status snapshot
Docker root	/var/lib/docker	Contains images, writable layers, logs and volumes
Public IP	212.47.78.250	Both production hostnames resolve here

Production paths

Purpose	Path
Solution Studio deployment	/home/renew/ai-effect-solution-studio-staging
German Portuguese and Danish repository	/home/renew/ai-effect-country-nodes/ai-effect-wp3-main
Dutch repository	/home/renew/ai-effect-country-nodes/ai-effect-wp3-dutch
German VILLAS source	/home/renew/ai-effect-country-nodes/villas-node
Production overlays and acceptance tools	/home/renew/ai-effect-country-nodes/overlays
WP3 orchestrator	/home/renew/ai-effect-wp3
Legacy portal retained for rollback	/home/renew/ai-effect-wp3-portal
Docker managed data	/var/lib/docker

Important Do not edit files directly under /var/lib/docker. Manage images, containers and volumes only through Docker and the approved Compose files.

4 Country node services and data

Node	Production containers	Persistent data	Acceptance result
Germany	data-provider, villas-chronics, villas-node, output-formatter	germany-node_shared-data	PASS: ProvideData, GenerateChronics and FormatOutput
Portugal	tef-data-provision, tef-knowledge-store, pt-synthetic-data, tef-clickhouse	ClickHouse data and synthetic models	PASS: ExecuteQuery, ApplyFunction, TrainModel and GenerateData
Denmark	dk-config-assistant, dk-topology-generator, dk-draw-topology, dk-ollama	Denmark data and Ollama model volumes	PASS: BuildRunConfig, GenerateTopology and RenderTopologies
Netherlands	nl-grid-synth-service, nl-benchmark-runner	Dutch synthetic-model volume	PASS: ConfigureAndSynthesize and RunBenchmark

Shared platform containers

- **Solution Studio** web, API and PostgreSQL containers.
- **WP3 execution** orchestrator API, Redis and three worker containers.
- **Ingress** the existing powerpal-traefik container terminates TLS and routes both AI EFFECT hostnames.
- **Legacy rollback** ai-effect-nginx is stopped; the legacy backend and database remain temporarily available for controlled rollback.

Data ownership

Git repositories and pinned image definitions make application code reproducible. PostgreSQL, ClickHouse and any user-generated workflow artifacts are stateful and require backups. Ollama models and most generated test data can be downloaded or rebuilt, but rebuilding them takes time and should not be treated as a substitute for a documented recovery procedure.

5 Capacity and storage

Current usage

Resource	Current measurement	Interpretation
Server disk	89 GiB used of 196 GiB	48 percent used; 97 GiB available
All Docker images	48.61 GB	Includes AI EFFECT, PowerPal and older EVPA projects
Reclaimable images	23.63 GB	Do not remove until rollback requirements are reviewed
Docker volumes	6.20 GB	Includes databases and other projects
Docker build cache	3.52 GB	Can be cleaned after release stabilization
AI EFFECT image estimate	Approximately 25 to 26 GB gross	Large images dominate; shared layers may lower true incremental use
AI EFFECT persistent volumes	Approximately 1.4 GB	Mostly Ollama and ClickHouse data
AI EFFECT operational budget	40 to 50 GB	Allows for upgrades, old images, logs and moderate data growth

Largest AI EFFECT components

Component	Approximate size
Dutch benchmark image	6.97 GB
Denmark Ollama image	5.57 GB
Denmark Ollama model volume	0.94 GB
Portuguese synthetic-data image	4.40 GB
German VILLAS image	3.08 GB
Dutch synthesizer image	0.95 GB
Portuguese ClickHouse volume	0.34 GB

Capacity thresholds

- **Disk warning** 70 percent used. Review images, logs, backups and database growth.
- **Disk critical** 80 percent used. Stop nonessential builds and add capacity or safely remove confirmed obsolete data.
- **Memory warning** 80 percent sustained for 15 minutes.
- **CPU warning** 85 percent sustained for 15 minutes, especially during Dutch benchmarking or Ollama execution.
- **Scale decision** Move AI EFFECT to a dedicated VPS before sustained public workloads, sensitive user data, or repeated resource contention.

6 Expenses and budget planning

All four country nodes run inside one fixed-price VPS. Docker does not charge per container, so the four nodes do not create four separate hosting bills. Cost changes occur only when the VPS plan, storage, backup services, domains, or external services change.

Cost item	Planning estimate	Notes
Existing Contabo VPS	Approximately EUR 14 per month	Matches the older Cloud VPS 30 shape. Confirm the actual invoice, VAT, region and billing term in the Contabo account.
AI EFFECT incremental compute cost	EUR 0 while sharing the paid VPS	AI EFFECT consumes capacity but does not add a per-container fee.
Optional object storage	From EUR 2.49 per month for 250 GB	Recommended for encrypted off-server database backups.
Comparable dedicated AI EFFECT VPS	Approximately EUR 14 per month	Would isolate AI EFFECT from PowerPal and EVPA workloads. Pricing changes over time.
Storage extension	Plan-specific	Not required at present; 97 GiB is available.
Domains DNS and email	Account-specific	Managed and billed separately from Docker containers.

Budget recommendation

For the current controlled release, retain the existing VPS and budget for off-server backup storage. For a production service with sustained external usage, budget for a separate AI EFFECT VPS of similar size plus backup storage. This removes competition with unrelated projects and reduces the security blast radius.

Billing caveat SSH access cannot reveal the contractual invoice. IT or finance must confirm the active Contabo product, renewal date, VAT treatment, add-ons and domain expenses in the relevant accounts.

7 Security controls

Controls already in place

- HTTPS and valid TLS certificates are provided by Traefik for both production hostnames.
- Country services use private Docker networking and publish no host ports.
- The Solution Studio web service binds locally to 127.0.0.1:13000.
- The WP3 orchestrator binds locally to 127.0.0.1:18000 and Redis has no public host port.
- Production environment files are stored on the server with mode 0600.
- SSH access uses individual public keys and the former developer keys were removed.
- Management pages and management API routes require HTTP Basic Authentication.
- The Portuguese production source excludes development secrets, a private test certificate, cached datasets and cached models.

Required security improvements

Priority	Action	Reason
Critical	Back up the new database off-server and test restoration	The active Solution Studio database is not covered by the current legacy backup job
High	Delete the temporary initial-credentials file after secure transfer and rotate the browser credential	Reduces credential exposure and prevents indefinite reuse
High	Review host and Contabo firewall rules	Only ports required by approved projects should be reachable from the internet
High	Add Docker log limits	Default json-file logging may grow without a bound
High	Move to application user accounts or institutional single sign-on	Basic Authentication is a temporary shared gate, not an auditable identity system
Medium	Apply Ubuntu and Docker security updates on a scheduled maintenance window	Reduces exposure to known vulnerabilities
Medium	Add container image vulnerability scanning and dependency review	Country images include large third-party runtimes
Medium	Separate AI EFFECT from unrelated workloads	Reduces resource contention and shared-host attack surface

Secret handling rules

- Never place passwords, API keys, private keys or environment files in Git, tickets, email or this document.
- Store production credentials in an institutional password manager with named-user access and an access log.
- Use separate credentials for production, staging and development.
- Rotate credentials after staff departure, suspected disclosure or handover to a new operations owner.
- Keep the SSH private key off the server and protect it with device encryption and a passphrase where supported.

8 Backup and recovery

Current backup state

A cron job runs at 02:30 each day and retains fourteen days of compressed PostgreSQL dumps. However, the job executes pg_dump inside ai-effect-postgres, which is the legacy portal database. It does not currently protect ai-effect-solution-studio-postgres-1, the active Solution Studio database.

Data set	Current state	Required recovery control
Solution Studio PostgreSQL	Not covered by the observed nightly job	Nightly pg_dump, 14 to 30 day retention, encrypted off-server copy and quarterly restore test
Legacy portal PostgreSQL	Nightly compressed dump with 14 day retention	Retain temporarily for rollback, then archive according to an approved retirement date
Portuguese ClickHouse	Approximately 342 MB in a Docker volume	Back up if data is authoritative; otherwise document the repeatable reconstruction procedure
Ollama model volume	Approximately 941 MB	Can be downloaded again; record the exact model name and version
Country repositories and overlays	Tracked in Git	Protect GitHub access and maintain pinned commits and release tags
Docker images	Stored locally and partly reproducible	Keep immutable image tags or registry copies for each production release
TLS and Traefik configuration	Stored on the shared server	Include configuration and certificate recovery procedure in infrastructure backup

Minimum recovery objectives

Objective	Initial target	Meaning
Recovery point objective	24 hours	At most one day of database changes may be lost
Recovery time objective	4 hours	Restore portal service within four hours during supported operations
Backup retention	14 to 30 days	Keep enough restore points to recover from delayed discovery of corruption
Restore validation	Quarterly and after backup changes	A backup is not accepted until it has been restored and checked

Restore test record

For every restore test, record the backup timestamp, operator, target environment, start and finish times, verification result, defects found and corrective actions. Restore tests must use an isolated database or disposable environment and must never overwrite production without an approved incident decision.

9 Monitoring and alerting

Monitor	Check	Suggested threshold	Response
Public availability	HTTPS GET to platform.ai-effect.eu	Two failures within five minutes	Check Traefik, DNS, web health and certificate status
API health	GET /api/health	Non-200 response or invalid payload	Check web proxy, API and PostgreSQL
Container health	Docker health and restart state	Unhealthy, exited or restart loop	Review logs and dependency health; restart only after identifying the cause
Disk	Root filesystem used percentage	Warning 70 percent; critical 80 percent	Review logs, images, build cache, backups and database growth
Memory	Available memory and swap activity	More than 80 percent used for 15 minutes	Identify high-memory workflow; apply limits or scale
CPU	Host and container CPU	More than 85 percent for 15 minutes	Check Dutch benchmark, Ollama and concurrent jobs
Backups	Age and exit status of newest backup	No successful backup within 26 hours	Treat as a production incident
TLS	Certificate expiry	Less than 21 days remaining	Investigate Traefik ACME renewal
Workflow reliability	Failed or stuck workflow count	Any repeated failure or task beyond agreed runtime	Inspect orchestrator and country-node logs

Logging requirements

- Configure per-container log rotation, for example a 10 to 50 MB maximum file size and three retained files.
- Do not log credentials, authorization headers, full environment variables or private workflow inputs.
- Retain security and administrative events long enough for investigation and governance requirements.
- Record deployment version, operator, timestamp and verification outcome for each production change.

10 Routine operations and change management

Daily checks

- Confirm the portal and API health endpoint respond successfully.
- Confirm production containers are running and healthy.
- Confirm the newest database backup completed and exists off-server.
- Review new critical alerts, restart loops and disk usage changes.

Weekly checks

- Review disk, image, volume and log growth.
- Review failed and long-running workflows.
- Review SSH access and recent production changes.
- Confirm DNS, TLS and Basic Authentication behavior on both hostnames.

Monthly checks

- Apply approved Ubuntu, Docker and application security updates during a maintenance window.
- Review container image vulnerabilities and pinned dependencies.
- Review Contabo usage, invoice, domain renewal status and backup storage consumption.
- Remove obsolete images only after confirming rollback requirements and successful backups.
- Review whether shared-host isolation remains acceptable.

Controlled production update

1. Record the change request, release commit, affected services, owner, risk and rollback version.
2. Confirm the Git working tree is clean and the release commit is pushed to the institutional repository.
3. Create and verify a fresh database backup and, for major updates, a Contabo snapshot.
4. Validate the Docker Compose configuration and confirm secret files retain mode 0600.
5. Build or pull immutable images. Avoid relying only on the latest tag for production releases.
6. Deploy the affected services without deleting persistent volumes.
7. Verify HTTPS, API health, public catalogue, protected management paths and container health.
8. Run the four-country acceptance validator when orchestration, service metadata or country images change.
9. Record the result and retain the previous image set until the agreed stabilization period ends.

Change rule Never run docker compose down with volume deletion flags in production unless a separately approved data-destruction procedure identifies the exact volumes and confirms recoverable backups.

11 Incident response and rollback

Initial response

1. Confirm the incident scope: one browser, one hostname, one container, one country node or the complete server.
2. Preserve evidence. Record timestamps, recent changes, container status and relevant logs before restarting services.
3. Protect data. Stop further workflow submissions if they could cause corruption or repeated failed writes.
4. Restore the smallest affected layer first: container, application release, database or full VPS.
5. Verify service health and business behavior, not only that a container is running.
6. Document the cause, recovery steps and preventive action.

Rollback options

Failure	Preferred rollback	Caution
New web or API image	Redeploy the previously recorded immutable image tag	Do not change database state unless migration compatibility is confirmed
Country-node image	Restore the previous pinned image and rerun that country acceptance workflow	Preserve shared volumes unless corruption is proven
Database migration	Use the documented application rollback or restore a verified pre-change dump	Database restore can discard newer data and requires incident approval
Traefik routing	Reapply the last known-good labels and verify both hostnames	Avoid exposing private API, orchestrator or country ports
Complete release failure	Temporarily re-enable the retained legacy frontend only under an approved rollback plan	The legacy service is not the long-term production target
Server loss	Provision a replacement VPS, restore configuration and databases, then repoint DNS	Requires off-server backups and documented infrastructure dependencies

12 Risks and prioritized action plan

Priority	Risk	Required action	Target
P0	Active database is not in the observed nightly backup	Create off-server Solution Studio backups and complete a restore test	Before accepting important user-created solutions
P1	German services and ClickHouse may not restart after a host reboot	Apply unless-stopped policies and Docker health checks	Before next planned reboot
P1	Docker json-file logs have no confirmed bound	Set daemon or per-service log rotation limits	Within one week
P1	Old evpa-nginx-1 is in a continuous restart loop	Identify owner, preserve necessary evidence, then repair or stop it	Immediate shared-host housekeeping
P1	Basic Authentication uses a shared credential	Rotate credential and plan named-user authentication or institutional SSO	Before wider team access
P2	AI EFFECT shares the server with unrelated projects and public ports	Review firewall and plan dedicated hosting when usage increases	Before sustained public production
P2	Large images consume disk and slow recovery	Create CPU-appropriate images where possible and retain immutable release tags	Next image optimization cycle
P2	No external alerting was verified	Implement uptime, disk, backup and container-health alerts	Before public announcement
P3	Legacy portal resources remain on the host	Set a retirement date after the rollback window and archive required data	After release stabilization

Release readiness decision

The platform is technically operational and has passed four-country functional acceptance. IT should treat backup coverage, restart behavior, log rotation and alerting as release-hardening work. The database backup gap is the only item that should prevent storing business-critical user data until corrected and tested.

13 Ownership and access management

Responsibility	Suggested owner	Record to maintain
Contabo account and invoices	Institutional IT or finance	Account owner, renewal date, billing contact, plan and add-ons
DNS and domains	Institutional IT	Registrar, expiry, DNS zone owner and recovery contact
SSH administration	Named IT administrators	Authorized key owner, issue date, review date and removal date
Production deployment	Application operations owner	Release commit, images, approver, deployment time and validation result
Database backups	Infrastructure operations	Backup age, off-server location, retention and restore-test evidence
Security response	Institutional security contact	Incident severity, evidence, decisions, notifications and remediation
Country-node functional ownership	Relevant node technical leads	Pinned source version, image version and acceptance result

Access review checklist

- Every administrator has an individual SSH key; shared private keys are prohibited.
- Departed or transferred staff are removed from authorized_keys and password-manager access.
- Production browser credentials are rotated and distributed only through the password manager.
- GitHub organization permissions and deployment branch protection are reviewed.
- Contabo, DNS and domain accounts use multifactor authentication and institutional recovery contacts.
- At least two authorized IT staff can recover the service without relying on one developer laptop.

Appendix A Operational commands

The following commands are read-only and are suitable for routine diagnosis after connecting to the server as the renew user. They do not display secret values.

Portal and API checks

```
curl -fsS -o /dev/null -w '%{http_code}\n' https://platform.ai-effect.eu/
curl -fsS https://platform.ai-effect.eu/api/health
curl -fsS -L https://platform.ai-effect.eu/api/services | head
```

Docker and capacity checks

```
docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}'
docker stats --no-stream
docker system df
df -hT /
free -h
```

Service logs

```
docker logs --tail 200 ai-effect-solution-studio-web-1
docker logs --tail 200 ai-effect-solution-studio-api-1
docker logs --tail 200 orchestrator-api-1
docker logs --tail 200 <country-container-name>
```

Four-country acceptance

```
cd /home/renew/ai-effect-country-nodes/overlays
python3 import_release_candidate.py
python3 validate_release_candidate.py
```

Operational caution Do not run prune, volume deletion, database restore, firewall modification or compose-down commands during diagnosis unless the exact impact and rollback path have been approved.

Appendix B Reference sources

Production acceptance record docs/PRODUCTION_ACCEPTANCE_2026-10-04.md in the Solution Studio repository

Previous developer handover Handover-to-Adarsh.md supplied to the project

Contabo VPS product information View official source

Contabo Cloud VPS documentation View official source

Contabo Object Storage pricing View official source

Contabo Auto Backup information View official source

Measurement notes

Capacity, container and storage values are point-in-time measurements from 4 October 2026. Docker image size totals may count shared layers differently from physical disk usage. Pricing is a planning estimate based on publicly listed Contabo products and must be reconciled with the actual institutional invoice.

Revision history

Version	Date	Change
1.0	4 October 2026	Initial IT operations and production management guide following Solution Studio cutover